

General Physics

Measurement

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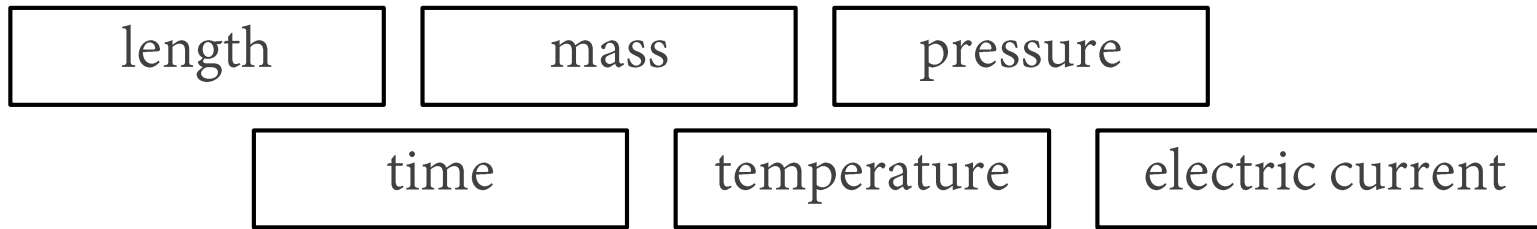
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1.1 Measuring things

- Science and engineering are based on precise measurements & comparisons of physical quantities
- Therefore, we need..
 - Rules: How are things measured? How are things compared?
 - Units: To represent the quantity of measurements
- One purpose of physics is to design and conduct experiments to establish rules and units

Physical quantities 物理量

When we conduct physics or engineering experiments, we need to measure several quantities...



We measure each physical quantity in its own units by comparing it with a “**standard**”

Unit 單位

- Unit
 - A unique name we assign to measures of that quantify
(所賦予某個物理量的度量的一個專門的名字)
- e.g., Length ↔ Meter (m)

quantity

unit

The “**standard**” being compared corresponds to exactly 1.0 unit of the quantity
(e.g., the standard for length = 1.0 m = distance traveled by light in a vacuum during a certain fraction of a second)

Base quantities

- There are so many physical quantities...
- They are not all independent (e.g. length vs. speed)
- Base quantities:
 - Some chosen physical quantities (7 quantities)

基本量	英文	SI 基本單位	單位符號
時間	Time	秒 (second)	s
長度	Length	公尺 (meter)	m
質量	Mass	公斤 (kilogram)	kg
電流	Electric current	安培 (ampere)	A
熱力學溫度	Thermodynamic temperature	克耳文 (kelvin)	K
物質的量	Amount of substance	莫耳 (mole)	mol
發光強度	Luminous intensity	燭光 (candela)	cd

Base quantities

- Base quantities
 - all have been assigned “standards”
 - be used to define all other physical quantities
 - e.g., speed is defined in terms of length (m) and time (s)
- Base standard
 - standard for base quantities
 - ✓ Accessible
 - ✓ Invariable: the measurement does not change over time

International System of Units

International System of Units (SI unit or metric system)

Table 1-1 Units for Three SI
Base Quantities

Quantity	Unit Name	Unit Symbol
Length	meter	m
Time	second	s
Mass	kilogram	kg

SI-derived units

Many SI-derived units are defined in terms of SI base units

- SI unit for energy
 - 1 Joules = 1 J = 1 kg m²/s²
- SI unit for power
 - 1 Watts = 1 W = 1 J/s = 1 kg m²/s³
(Kilogram-meter squared per second cubed)

Scientific notation 科學符號

- To express very large and very small quantities, we use scientific notation
- It employs power of 10 to write a number,
- e.g.,

$$3\,560\,000\,000\text{ m} = 3.56 \times 10^9\text{ m}$$

$$0.000\,000\,492\text{ s} = 4.92 \times 10^{-7}\text{ s.}$$

In computers, the power of 10 is expressed by E

$$3.56 \times 10^9\text{ m} \longrightarrow 3.56\text{E}9\text{ m}$$

$$4.92 \times 10^{-7}\text{ s.} \longrightarrow 4.92\text{E} - 7\text{ s}$$

Prefixes 字首

Sometimes, we use prefixes to represent a certain power of 10

Table 1-2 Prefixes for SI Units

Factor	Prefix ^a	Symbol
10 ²⁴	yotta-	Y
10 ²¹	zetta-	Z
10 ¹⁸	exa-	E
10 ¹⁵	peta-	P
10 ¹²	tera-	T
10⁹	giga-	G
10⁶	mega-	M
10³	kilo-	k
10 ²	hecto-	h
10 ¹	deka-	da
10 ⁻¹	deci-	d
10⁻²	centi-	c
10⁻³	milli-	m
10⁻⁶	micro-	μ
10⁻⁹	nano-	n
10⁻¹²	pico-	p
10 ⁻¹⁵	femto-	f
10 ⁻¹⁸	atto-	a
10 ⁻²¹	zepto-	z
10 ⁻²⁴	yocto-	y

→ $1.27 \times 10^9 \text{ watts} = 1.27 \text{ gigawatts} = 1.27 \text{ GW}$

→ $2.35 \times 10^{-9} \text{ s} = 2.35 \text{ nanoseconds} = 2.35 \text{ ns}$

^aThe most frequently used prefixes are shown in bold type.

Changing unit

- We use a method called “chain-link conversion” to change the unit
 - Multiply the original quantity by a “conversion factor” (轉換因子)
- Conversion factor
 - A ratio of units that is equal to 1
 - e.g., $\frac{60\text{ s}}{1\text{ min}}$
(involve numbers and units)
- e.g., convert 2 min to seconds

$$2\text{ min} = (2\text{ min})(1) = (2\cancel{\text{ min}})\left(\frac{60\text{ s}}{1\cancel{\text{ min}}}\right) = 120\text{ s}.$$

Practice - 01

Chain-link conversions:

- $1.3 \text{ km} \rightarrow ? \text{ m}$
- $0.8 \text{ km} \rightarrow ? \text{ cm}$
- $2845 \text{ mm} \rightarrow ? \text{ ft}$

Length

- The standard for the unit of length has been defined by:
 - One ten-millionth (千萬分之一) of the distance from the North Pole to the equator
 - A platinum-iridium (鉑銥合金) standard meter bar kept in France
 - 1650763.73 wavelengths of an orange-red light emitted by atoms of krypton-86 (氪86)
- Today



The meter is the length of the path traveled by light in a vacuum during a time interval of $1/299\,792\,458$ of a second.

Significant figure 有效位數

- Significant figures are meaningful digits
- The quantity is often rounded to match the fewest number of significant figures

e.g., 11.3516 is rounded to 3 significant figures

$$11.3516 \rightarrow 11.4$$

e.g., 11.3279 is rounded to 3 significant figures

$$11.3279 \rightarrow 11.3$$

Some rules for significant figures

- The significant figures are the digits that carry meaning, contributing to the measurement resolution
- All non-zero digits are considered significant. For example, 91 has two significant figures (9 and 1), while 123.45 has five significant figures (1, 2, 3, 4, and 5)
- Zeros appearing anywhere between two non-zero digits are significant. e.g., 101.1203 has seven significant figures: 1, 0, 1, 1, 2, 0, and 3
- Leading zeros are not significant. e.g. 0.00052 has two significant figures: 5 and 2
- Trailing zeros in a number containing a decimal point are significant. e.g., 12.2300 has six significant figures: 1, 2, 2, 3, 0, and 0

Sample problem 1.1.1

The largest ball of string in the world is about 2 m in radius. To the nearest order of magnitude, what is the total length L of the string in the ball? Assume the cross-section of the string is square with an edge length $d = 4$ mm.



1.2 Time

- In scientific works, we want to know how long an event lasts.
- Therefore, any time standard must be able to answer two questions
 - When did it happen?
 - What is its duration?

Table 1-4 Some Approximate Time Intervals

Measurement	Time Interval in Seconds	Measurement	Time Interval in Seconds
Lifetime of the proton (predicted)	3×10^{40}	Time between human heartbeats	8×10^{-1}
Age of the universe	5×10^{17}	Lifetime of the muon	2×10^{-6}
Age of the pyramid of Cheops	1×10^{11}	Shortest lab light pulse	1×10^{-16}
Human life expectancy	2×10^9	Lifetime of the most unstable particle	1×10^{-23}
Length of a day	9×10^4	The Planck time ^a	1×10^{-43}

Standard of time

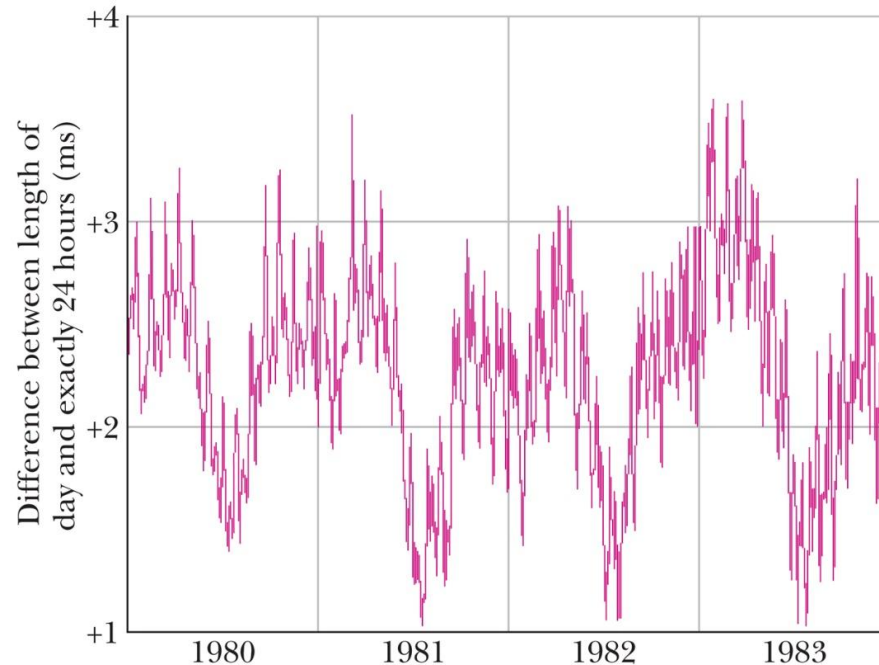
- Rotation of Earth
 - determine the length of a day
- Quartz clocks
 - quartz vibration continuously
- Atomic clocks



One second is the time taken by 9 192 631 770 oscillations of the light (of a specified wavelength) emitted by a cesium-133 atom.

Variation in the length of a day

- The variation in the length of a day as measured by an atomic clock
- Atomic clock is consistent: 2 cesium clocks would have to run for 6000 years before their readings would differ by more than 1 s



This shows the precision of atomic clocks and the relative imprecision of Earth's rotation

Sample problem 1.2.1

Light-fermi is the time taken by light to travel a distance of 1 fermi (1 fermi = 1 femtometer = 1 fm).
How many seconds are in a light-fermi?

Practice - 02

The height of a motion picture film is 35.0 cm. If 24 frames go by in 1.0 s, calculate the total number of frames required to show a 2.0 h long motion picture.

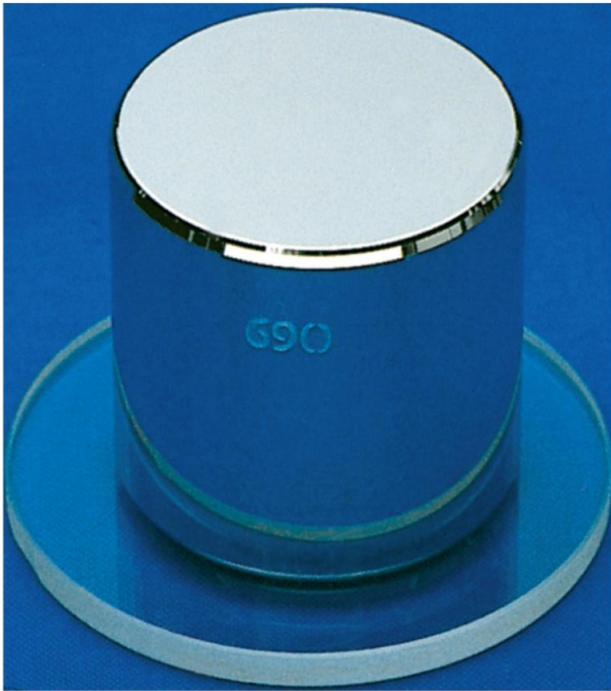
Practice - 03

The age of the universe is approximately 10^{10} years, and mankind has existed for about 10^6 years. If the age of the universe were 1.0 day, how many seconds would mankind have existed?

1-3 Mass

The SI standard of mass is a cylinder of platinum (鉑) and iridium (銱) stored in France

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- A mass of one kilogram
- Accurate copies were sent around the world; other masses can be measured by comparing them against these copies

Second mass standard

The atomic mass unit (u) is a second mass standard

- 1 atom of Carbon-12 is assigned a mass of 12 u
- $1 \text{ u} = 1.66053886 \times 10^{-27} \text{ kg}$ ($\pm 10 \times 10^{-35} \text{ kg}$)
- Used for measuring the masses of atoms and molecules

Density

Density (ρ) is defined as the mass per unit volume:
(發音: rho)

$$\rho = \frac{m}{V}$$

e.g.,

- Density of material: $(18 \text{ kg}) / (0.032 \text{ m}^3) = 560 \text{ kg/m}^3$
- Mass of object: $(380 \text{ kg/m}^3) \times (0.0040 \text{ m}^3) = 1.5 \text{ kg}$
- Volume of object: $(250 \text{ kg}) / (1280 \text{ kg/m}^3) = 0.20 \text{ m}^3$

Sample problem 1.3.1

A heavy object can sink into the ground during an earthquake if the shaking causes the ground to undergo liquefaction, in which the soil grains experience little friction as they slide over one another. The ground is then effectively quicksand. The possibility of liquefaction in sandy ground can be predicted in terms of the void ratio e for a sample of the ground

$$e = \frac{V_{voids}}{V_{grains}}$$

Here, V_{grains} is the total volume of the sand grains in the sample, and V_{voids} is the total volume between the grains. If e exceeds a critical value of 0.8, liquefaction can occur during an earthquake. What is the corresponding sand density r_{sand} ? Solid silicon dioxide (the primary component of sand) has a density of $2.6 \times 10^3 \text{ kg/m}^3$



Sample problem 1.3.1

Practice - 04

Gold has a density of 19.32 g/cm^3 .

- (a) If a sample of gold with a mass of 29.34 g is pressed into a leaf of $1.000 \mu\text{m}$ thickness, what is the area of the leaf?
- (b) If the gold is drawn out into a cylindrical fiber of radius $2.5 \mu\text{m}$, what is the length of the fiber?